

Resonant: A Web-Based Extended Reality Platform for Heartbeat Sharing via Headset Ballistocardiography and Mobile Remote Photoplethysmography

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ABSTRACT

This paper presents Resonant, a work-in-progress Web-based Extended Reality platform designed to capture and share cardiac signals through haptics using virtual reality headsets and mobile phones. The system integrates headset-based ballistocardiography, mobile remote photoplethysmography, and commercial heart rate monitors for validation and comparison. This system enables heart rate monitoring without the need for an external heart rate device, as well as provides haptic feedback. Resonant is motivated by the goal of enabling physiological signal sharing without requiring specialized sensing devices to support novel XR experiences that can enable better social connections. This paper describes the system design, sensing approaches, and a workflow for synchronizing heartbeat-driven haptic feedback with narrative media. The intended psychophysiological effects remain to be evaluated in future studies.

Index Terms: Extended Reality, Haptics, Biofeedback, Physiological Computing, Empathy, rPPG, Ballistocardiography, HRV, Work in Progress

1 INTRODUCTION

Extended Reality systems offer rich visual and auditory experiences, yet they rarely include access to internal physiological signals such as heart rate or heart rate variability. As a result, many mediated experiences lack information about a user's internal state, particularly in social, narrative, and therapeutic settings. Physiological signals can reveal changes in arousal, stress, and emotion that are not always visible through speech or facial expression alone.

Previous work has shown that perceiving another person's heartbeat can increase empathy and feelings of social closeness [14, 6], as well as influence social cognition [2]. Haptic representations of physiological signals have also been found to shape emotional experience, even when the feedback does not correspond to the user's actual physiological state [3]. Despite these benefits, effective physiological sensing and haptic feedback systems typically rely on additional hardware, which introduces friction and limits their use outside controlled research settings.

Resonant is motivated by a desire to reduce reliance on specialized physiological hardware and to support the sharing of physiological signals using devices that people already own. Modern smartphones and head-mounted displays include cameras, inertial

sensors, and vibration motors that can be repurposed for physiological sensing and haptic feedback. By combining these capabilities, Resonant explores how cardiac signals can be sensed, rendered, and shared within Extended Reality environments.

This paper introduces Resonant as a work-in-progress system. The primary contributions are:

1. Integration of headset-based ballistocardiography and mobile remote photoplethysmography into a cross modal system for heartbeat estimation and transfer across sensing modalities.
2. A unified haptic delivery pipeline that supports heartbeat-driven feedback across multiple devices and form factors in Extended Reality experiences.
3. A heartbeat-synchronized narrative workflow that aligns cardiac events with video playback using a subtitle-style representation referred to as Heartbeat SRT.

2 BACKGROUND AND MOTIVATION

2.1 Physiological Sensing in Extended Reality

Physiological computing has traditionally relied on dedicated sensing devices for emotion and state monitoring, including electrocardiography, photoplethysmography, and electroencephalography. While these devices can provide accurate measurements, they introduce additional setup requirements, require direct physical contact, and reduce accessibility outside laboratory environments. As a result, many physiological Extended Reality experiences remain confined to controlled research settings.

Consumer devices such as smartwatches have increased the availability of physiological sensing in everyday contexts by integrating optical heart rate sensors into wearable form factors. However, these systems still require additional hardware, ongoing device management, and user compliance, which can limit their practicality for spontaneous or large scale XR deployments.

Recent work has explored sensorless and contactless approaches that leverage sensors already embedded in consumer hardware. Hernandez et al. demonstrated that inertial sensors in head-mounted wearables can capture cardiac-induced micro-movements using ballistocardiography [5]. Subsequent studies validated heart rate extraction from VR headset motion under controlled conditions [11]. More recently, Roupia et al. introduced a spectral framework that captures both cardiac and respiratory rates from subtle VR device movements [10].

Resonant builds on this body of work by incorporating headset-based ballistocardiography (BCG) and mobile remote photoplethysmography (rPPG) into an interactive XR system. The goal is not to replace clinical sensing devices but to provide a practical and scalable means of estimating cardiac signals for novel social applications. Even though these systems have limitations with regard to their accuracy in certain situations, such as high movement, we believe that by adding this option, it enables more access to unique social experiences.

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2.2 Remote Photoplethysmography

Remote photoplethysmography estimates cardiac activity by analyzing subtle color variations in skin caused by blood volume changes. rPPG enables camera-based physiological sensing using cameras and has become a promising alternative to dedicated hardware.

Resonant supports two categories of rPPG methods. First, it implements a Plane-Orthogonal-to-Skin based pipeline following the algorithmic principles described by Wang et al. [13]. This signal processing approach projects normalized RGB signals onto a chrominance subspace to reduce illumination artifacts and isolate the pulsatile component. The implementation is based on rPPG-Toolbox [7] and allows for a quicker algorithmic method for real time heart rate capture.

In addition to POS-based rPPG, Resonant includes a deep learning option based on the memory-efficient temporal-spatial state space duality approach proposed by Wang et al. [12]. This method is designed to reduce memory usage and inference latency while maintaining robust estimation and uses the open-rppg library. Ongoing work includes integrating additional deep learning models documented in the rPPG-Toolbox [7] as well as open-rppg and comparing their behavior across different capture conditions. All rPPG processing is performed within a backend Python (Flask) pipeline.

2.3 Haptics, Interoception, and Empathy

Haptic feedback provides a direct channel for communicating physiological signals. Vibrotactile heartbeat feedback has been shown to foster empathy and social connection in mediated interaction [6, 4]. Synchronizing stimuli with cardiac signals can also influence social perception and bias [2].

Beyond social effects, altered cardiac feedback has been shown to modulate emotional experience, suggesting a bottom-up pathway for affective regulation [3]. Most prior work represents the heartbeat as a regular rhythm, overlooking Heart Rate Variability. HRV reflects autonomic regulation and is associated with emotional and social functioning [1, 9]. Resonant aims to explore whether HRV can be meaningfully conveyed through simple haptic feedback.

3 SYSTEM ARCHITECTURE

Resonant consists of a WebXR frontend, a backend Python processing pipeline, and haptic rendering on consumer devices. The system uses WebXR through the A-Frame framework to construct XR scenes, handle interaction, and manage video playback. A-Frame was chosen because it simplifies communication and synchronization between mobile and headset components, allowing the system to support cross-device heartbeat sharing. The movement and video data are captured on-device, transmitted to the backend for analysis, and returned as heart beat and heartbeat variability streams. No physiological signal processing or machine learning inference is performed on the client.

3.1 Headset Ballistocardiography

When a VR headset is available, cardiac signals are estimated from inertial motion data by processing the headset's embedded accelerometer and gyroscope streams [10]. The pipeline first filters and normalizes the raw motion data to reduce the influence of large voluntary head movements, then projects the remaining signal onto the axis that maximizes the tiny periodic recoil associated with each heartbeat [10]. A one dimensional Kalman filter is used conceptually to track the underlying rhythmic motion over time and a Fast Fourier Transform is applied on short windows of this filtered signal to identify the dominant frequency component which is converted to beats per minute for continuous cardiac rate estimation [10]. The current system focuses on cardiac rate estimation and treats respiratory signal analysis and haptic rendering as planned extensions in future work.

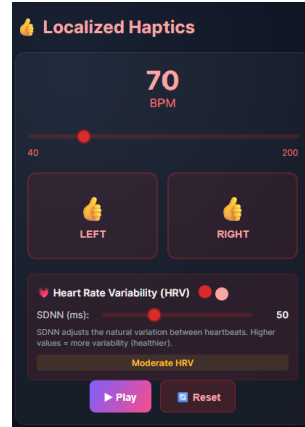


Figure 1: Example of Haptic Rendering on mobile device with Heart Rate Variability Changes and button placement

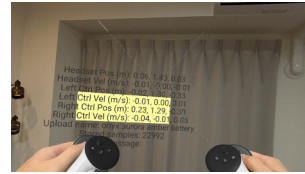


Figure 2: Example of recording haptic data through an A-Frame WebXR application

3.2 Mobile Remote Photoplethysmography

For users without access to a VR headset or wearable sensor, Resonant estimates cardiac activity from smartphone video using remote photoplethysmography. The system implements the classical Plane Orthogonal to Skin algorithm from the rPPG Toolbox [7, 13] as well as memory efficient deep learning models adapted from recent state space formulations through the open-rppg library [12]. The modular architecture allows different rPPG algorithms to be compared without changes to the client side experience and the team is currently testing additional models to expand the range of supported estimation methods across diverse lighting conditions and skin tones.

3.3 Commercial Heart Rate Monitors

Resonant integrates commercial heart rate monitors to provide reference measurements for validation and comparison. The current implementation supports the Polar H10 chest strap, which provides high-quality cardiac measurements commonly used in research settings. In addition to Polar H10, ongoing work includes integration with smartwatch platforms such as Apple Watch and other consumer heart rate devices. This allows for a baseline comparison as well as easy fine-tuning of the BCG algorithm, as you can collect data side by side.

3.4 Haptic Rendering

Heartbeat signals are delivered using standard vibration capabilities on consumer devices. On mobile platforms, the system uses the JavaScript Vibration API, while on WebXR headsets it relies on A-Frame vibration components. Instead of a single, uniform vibration, the haptic feedback varies the timing and duration of each pulse to approximate the two phases of the heartbeat. The initial systolic contraction (lub) is rendered as a shorter pulse, followed by a longer diastolic relaxation phase (dub), giving the user a sense

of the natural rhythm of the heart [6]. This method allows users to perceive the timing of individual heartbeats rather than just a steady metronome-like pulse.

The rendering pipeline supports multiple configuration modes for experimental flexibility. Heart Rate Variability (HRV) can be explicitly specified to introduce natural beat-to-beat interval variability, enabling investigation into whether users can distinguish high variability from low variability states and whether entrainment techniques might increase a user's HRV through prolonged haptic exposure [1]. Additionally, the system allows designation of specific anatomical zones on the device where haptic stimuli are localized, such as defined button regions where users place their fingers. Prior research suggests that fingertip regions exhibit heightened haptic sensitivity [8], making it valuable to empirically test whether localized haptic rendering produces stronger perceptual and emotional effects compared to diffuse device vibration. This can be seen in Fig. 1. Rendering of respiratory signals and combined cardiorespiratory haptics is planned for future work.

4 HEARTBEAT SRT

Resonant includes a workflow for synchronizing heartbeat-driven haptic feedback with narrative video. The goal is to associate physiological signals from a recorded speaker with the video timeline, enabling viewers to feel the speaker's cardiac activity during playback.

First, rPPG is applied to pre-recorded video to extract the blood volume pulse signal, from which individual heartbeat timings are identified. These timings are stored in a plain text file with entries containing timestamps and heartbeat interval data, following a format analogous to subtitle files. The file is generated offline and stored alongside the video. Second, during playback, the Heartbeat SRT file is read in parallel with the video. As the timeline advances, the system triggers corresponding haptic pulses on the viewer's device, maintaining temporal alignment even if playback is paused or otherwise disrupted.

An example Heartbeat SRT entry appears as follows:

```
1
00:00:15,200 --> 00:00:15,750
BPM: 82 | Interval: 730ms

2
00:00:15,750 --> 00:00:16,470
BPM: 82 | Interval: 720ms
```

The intended outcome is to allow viewers to feel physiological changes during emotionally significant moments in a narrative. This approach provides a lightweight method for augmenting video with cardiac haptics. The experiential impact has not yet been empirically evaluated and remains a topic for future user studies.

5 FUTURE RESEARCH DIRECTIONS AND POTENTIAL IMPACT

Resonant is designed to support future research across psychophysiology, affective computing, and experiential design. One immediate direction is the use of haptic cardiac feedback for studying entrainment-based anxiety regulation. Prior work has shown that altered or false cardiac feedback can reduce anxiety during stressful tasks by influencing perceived physiological state [3]. Resonant builds on this work by allowing controlled manipulation of both heart rate and Heart Rate Variability through haptic feedback. We propose an order-independent user study comparing user responses to watching a video under three conditions: (a) no haptics, (b) with haptics corresponding to the video, and (c) with fake haptics with no correlation with the video, where the heart rate is significantly off from the true heart rate. The study will explore whether guiding

users toward a specific heart rate or higher HRV leads to changes in self-reported anxiety and physiological arousal. This focus on HRV is particularly relevant, as HRV is closely linked to parasympathetic activity and emotional regulation [1, 9], yet has received limited attention in haptic biofeedback research.

A second research direction involves interpersonal physiological sharing in mediated environments. Previous studies have shown that perceiving another person's heartbeat can increase empathy, strengthen feelings of social connection, and influence social cognition [2, 6, 4]. Resonant enables this line of research without requiring participants to wear dedicated heart rate sensors, lowering barriers that have traditionally confined such studies to laboratory settings. By supporting multiple sensing modalities, including headset-based ballistocardiography and mobile remote photoplethysmography, the system allows cardiac signals to be shared between users in different physical and technological contexts. This makes it possible to study how shared physiological awareness affects collaboration, emotional alignment, and social dynamics in distributed XR environments.

A third direction concerns narrative and media experiences. The Heartbeat SRT workflow provides a mechanism for embedding cardiac timing data alongside video content, enabling viewers to feel the physiological state of a speaker or character during key narrative moments. This adds an interoceptive layer to storytelling that goes beyond traditional audiovisual presentation. Future work will examine whether synchronizing haptic feedback with a speaker's recorded cardiac activity influences viewer engagement, emotional response, or perceived authenticity in contexts such as interviews, testimony, or performance. Because Heartbeat SRT relies on a lightweight text-based format, these files can be distributed alongside standard video without specialized playback infrastructure, supporting potential use in documentary media, education, and digital humanities.

Building on this early technical prototype, future work will focus on empirically testing these hypotheses and using the system as a foundation for broader exploration of physiological interaction in immersive experiences.

6 CONCLUSION

Resonant is a work-in-progress platform for sharing cardiac signals in Extended Reality using commodity hardware. By combining headset-based ballistocardiography, mobile remote photoplethysmography, and commercial heart rate monitors, the system enables exploration of physiological haptics without dedicated sensing devices. This paper describes the system design and motivation, with future work focused on empirical evaluation of experiential and psychophysiological outcomes.

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